



Programming Fundamentals

LAB # 6

B8

WEEK 6

Monday

Instructions:

- You have to submit CPP files of the program in a folder, named your Registration Number.
- You have to upload your lab tasks on CMS.
- All program codes should be written in C/C++. Students should use Visual studio compiler for coding.
- Indent and comment your code.
- Use meaningful variable names
- Plan your code carefully on a piece of paper before you implement it.

Learning Objectives:

- Demonstrate knowledge of **functions** programming C++.
-

Write a C++ program that uses a **function with a call by value parameter and a return type** to:

- Take an integer input from the user.
- Return true if the number is prime, false otherwise.
- Display the result in main().

Expected output :

Enter a number: 13

13 is a prime number.

Write a C++ program that uses a **function with a call by value parameter (character array) and a return type** to:

- Read a character array (C-style string) from the user.
- Count and return the number of vowels (a, e, i, o, u - both upper and lower case) in the array.
- Display the count in main().

Expected Output:

Enter a word or sentence: Education

Number of vowels: 5

Write a C++ program that uses a **function with a call by value parameter and a return type**:

- Takes a positive integer N as input from the user.
- Calculates and returns the sum of all integers from 1 to N using a loop.
- Display the returned sum in `main()`

Expected Output:

Enter a positive integer: 5

The sum of numbers from 1 to 5 is: 15

Write a C++ program that contains **three functions other than main**.

- The first function should read a number n (as a decimal or whole number) and an exponent p (which can be positive or negative).
 - The second function should **manually calculate n raised to the power p** using loops (without using any built-in functions like `pow()`).
 - The third function should display the result.
- Your program should correctly handle both **positive and negative powers**.

Example Output 1:

Enter a number: 2

Enter the power (can be negative): -2

2 raised to the power -2 is: 0.25

Example Output 2:

Enter a number: 3

Enter the power (can be negative): 3

3 raised to the power 3 is: 27

Write a C++ program that contains **three functions other than main**:

- The **first function** reads an array of integer values from the user.
- The **second function** calculates the **sum** of the values in the array.
- The **third function** prints the **sum** calculated by the second function.

Sample Output:

Enter 5 integers:

10

20

30

40

50

The sum of array elements is: 150

Write a C++ program using **functions** and **file handling**.

Your program must contain **four functions (excluding main)**, where:

- *One function reads an integer from the user using `cin`.*
- *One function calculates the **cube (3rd power)** of the number.*
- *One function displays the result on the screen and also writes it to a file named `output.txt`.*
- *One function controls the flow of the program by calling the other three in the correct order.*

All functions must be written with **no return type** (void) and **no parameters**.

The `main()` function should **not contain any logic**, input/output statements, or calculations. It should only call the controller function.

Expected Output:

Console Output:

Enter a number: 5

The cube of 5 is: 125

Result written to output.txt

Contents of `output.txt`:

The cube of 5 is: 125

Write a C++ program that demonstrates **function calling inside another function**.

The `main()` function should call a function named `processInfo()`, which in turn calls another function named `readAndDisplay()`.

The `readAndDisplay()` function should:

- Read your **name** and **age** as input.
- Display your **name** and **age** on the screen.

Expected Output:

Enter your name: Qaiser Nadeem

Enter your age: 24

Your name is: Qaiser Nadeem

Your age is: 24

Hint:

- Use `getline(cin, name);` to read the full name (including spaces).
- Use `cin >> age;` to read the age.
- Make sure to call `readAndDisplay()` inside `processInfo()`.

End of LAB 6 😊